

Methods in Yield Aware

The Latent TTO is now built and complete and generated designs that are physically constraint. To build the Yield Aware optimization - the initial deliverable is defined below.

Deliverable - The Yield Aware Optimization is used to search, measure, and sort the designs that considering the yield in factory (I am using IPC to consider the factory deviation/wobble) and the optimizer actively hunts for a geometry that survives the factory wobble and ranks the designs based on the yield.

Try 1: Lambda 0 - no yield optimization - check with inverse model generated design.

```
=====
SAMPLE 1260 lambda_f=0.0 curriculum=False
=====

TOLERANCE MODEL [ipc] x scale 1 (1-sigma perturbations; limit=3*sigma Cpk=1 convention for 'ipc'):
VIA_RADIUS : abs 0.5 mil (limit/3)      sigma_phys=0.5 mil      sigma_norm=0.1873
ANTIPAD_RADIUS : abs 0.67 mil (limit/3)  sigma_phys=0.67 mil      sigma_norm=0.1287
PITCH : abs 0.33 mil (limit/3)           sigma_phys=0.33 mil      sigma_norm=0.0386
TDIEL : rel +/-3.3% (limit/3)            sigma_phys=0.9874 feat-units sigma_norm=0.0413
TMET : rel +/-3.3% (limit/3)            sigma_phys=0.08881 feat-units sigma_norm=0.0899
PERMITTIVITY : rel +/-0.67% (limit/3)     sigma_phys=0.02094 feat-units sigma_norm=0.0326
CONDUCTIVITY : rel +/-3.3% (limit/3)      sigma_phys=0.01423 decades sigma_norm=0.3035
LOSS/TANGENT : rel +/-3.3% (limit/3)      sigma_phys=0.01423 decades sigma_norm=0.0351
(IPC gives ACCEPTANCE LIMITS, not process sigmas; limit/3 is a stated Cpk=1 assumption. CONDUCTIVITY entry is an engineering assumption with no standards anchor. See PartC_Yield_Plan_v3.md.)

=== YIELD TTO [var_L0_ipc] sample 1260: 12 restarts x 150 steps
lambda_j=0.0 [ABLATION: variance proxy]
spec: eye-band IL >= target-1.0 dB, RL <= -8.0 dB | tol_frac=0.1
restart 00 | eye-fit 0.53 dB | sens 0.00356 | yield 11.2% [9.9,12.7] | max|x| 1.31
restart 01 | eye-fit 0.54 dB | sens 0.00368 | yield 9.7% [8.4,11.0] | max|x| 1.37
restart 02 | eye-fit 0.53 dB | sens 0.00350 | yield 10.8% [9.6,12.3] | max|x| 1.27
restart 03 | eye-fit 0.53 dB | sens 0.00350 | yield 13.2% [11.8,14.8] | max|x| 1.27
restart 04 | eye-fit 0.53 dB | sens 0.00354 | yield 10.9% [9.7,12.4] | max|x| 1.29
restart 05 | eye-fit 0.53 dB | sens 0.00349 | yield 12.8% [11.5,14.4] | max|x| 1.27
restart 06 | eye-fit 0.53 dB | sens 0.00348 | yield 10.8% [9.5,12.2] | max|x| 1.27
restart 07 | eye-fit 0.53 dB | sens 0.00348 | yield 11.0% [9.7,12.4] | max|x| 1.28
restart 08 | eye-fit 0.53 dB | sens 0.00348 | yield 11.7% [10.3,13.1] | max|x| 1.27
restart 09 | eye-fit 0.53 dB | sens 0.00350 | yield 10.4% [9.2,11.9] | max|x| 1.27
restart 10 | eye-fit 0.53 dB | sens 0.00348 | yield 9.6% [8.3,10.9] | max|x| 1.27
restart 11 | eye-fit 0.53 dB | sens 0.00350 | yield 12.8% [11.4,14.3] | max|x| 1.27

fit gate (1.5 dB eye-band): 12/12 candidates pass
portfolio diversity: pairwise |dx| mean 0.074, max 0.278 (normalized units; ~0 would mean all restarts found the same design)

=====
RANKED PORTFOLIO (top 5) sample 1260 [var_L0_ipc]
=====
rank eye-fit dB yield % sens file
1 0.53 13.2 0.00350 design_sample_1260_var_L0_ipc_r03.npz
2 0.53 12.8 0.00349 design_sample_1260_var_L0_ipc_r05.npz
3 0.53 12.8 0.00350 design_sample_1260_var_L0_ipc_r11.npz
4 0.53 11.7 0.00348 design_sample_1260_var_L0_ipc_r08.npz
5 0.53 11.2 0.00356 design_sample_1260_var_L0_ipc_r00.npz

csv : portfolio_sample_1260_var_L0_ipc_20260729_211010.csv
pareto : pareto_sample_1260_var_L0_ipc_20260729_211010.png
top design (for stage 07): design_sample_1260_var_L0_ipc_top.npz
```

fig1: direct fit - close match with no yield push

For the sample 1260 (one from validation dataset), the fit-only design with lambda 0 (yield Push is off) - here it is clear that the cVAE model chase the design with best fit (0.53dB) and passes the gate. But with no yield optimization (just yield measure here with lambda =0) the manufactured boards with design from this model will fail in the factory with slight wobbles with a yield of only 13.2% . This establishes the fit-only baseline that yield-aware optimization must improve upon.

```
=====
Stage 07: design_sample_1260_var_l0_ipc_top (template sim_pkg_1027, pair 1, geometry='x_local_norm')
=====
```

```
--- geometry sanity check [design_sample_1260_var_l0_ipc_top] ---
via_radius    = 12.213 mil
antipad_rad   = 20.577 mil
pitch         = 43.909 mil
TDIEL/TMET    = 177.4 total, eps_r=3.018, tan_d=0.0146
n_ports       = 12
geometry looks physical (via < antipad < pitch/2, eps_r>1).
```

```
[--check-only] stopping before simulation. If the geometry looks good, re-run without --check-only.
```

Fig2: Geometry of Top design with 13.2% yield

The near perfect design with matching the exact targets will leave no margin for the model and hence making it fragile with slight manufacturing variance.

Try 2: with Jacobian (proposed in the initial plan) Lambda = 1.0

```
=====
SAMPLE 1260 lambda_j=1.0 curriculum=False
=====

TOLERANCE MODEL [ipc] x scale 1 (1-sigma perturbations; limit=3*sigma Cpk=1 convention for 'ipc'):
VIA_RADIUS : abs 0.5 mil (limit/3)   sigma_phys=0.5 mil   sigma_norm=0.1073
ANTIPAD_RADIUS : abs 0.67 mil (limit/3) sigma_phys=0.67 mil   sigma_norm=0.1207
PITCH : abs 0.33 mil (limit/3)       sigma_phys=0.33 mil   sigma_norm=0.0386
TDIEL : rel +/-3.3% (limit/3)        sigma_phys=0.9074 feat-units sigma_norm=0.0413
TMET : rel +/-3.3% (limit/3)        sigma_phys=0.08801 feat-units sigma_norm=0.0899
PERMITTIVITY : rel +/-0.67% (limit/3) sigma_phys=0.02094 feat-units sigma_norm=0.0326
CONDUCTIVITY : rel +/-3.3% (limit/3) sigma_phys=0.01423 decades sigma_norm=0.3035
LOSSTANGENT : rel +/-3.3% (limit/3) sigma_phys=0.01423 decades sigma_norm=0.0351
(IPC gives ACCEPTANCE LIMITS, not process sigmas; limit/3 is a stated Cpk=1 assumption. CONDUCTIVITY entry is an engineering assumption with no standards anchor. See )

=== YIELD T10 [var_l1_ipc] sample 1260: 12 restarts x 150 steps
lambda_j=1.0 [ABLATION: variance proxy]
spec: eye-band IL >= target-1.0 dB, RL <= -8.0 dB | tol_frac=0.1
restart 00 | eye-fit 0.53 dB | sens 0.00327 | yield 11.2% [9.9,12.7] | max|x| 1.16
restart 01 | eye-fit 0.53 dB | sens 0.00336 | yield 9.8% [8.6,11.2] | max|x| 1.21
restart 02 | eye-fit 0.53 dB | sens 0.00322 | yield 10.8% [9.5,12.2] | max|x| 1.12
restart 03 | eye-fit 0.53 dB | sens 0.00326 | yield 13.2% [11.7,14.7] | max|x| 1.12
restart 04 | eye-fit 0.53 dB | sens 0.00326 | yield 11.1% [9.8,12.6] | max|x| 1.14
restart 05 | eye-fit 0.53 dB | sens 0.00324 | yield 13.1% [11.7,14.6] | max|x| 1.12
restart 06 | eye-fit 0.53 dB | sens 0.00324 | yield 10.8% [9.6,12.3] | max|x| 1.12
restart 07 | eye-fit 0.53 dB | sens 0.00325 | yield 10.9% [9.6,12.3] | max|x| 1.13
restart 08 | eye-fit 0.53 dB | sens 0.00322 | yield 11.7% [10.4,13.2] | max|x| 1.12
restart 09 | eye-fit 0.53 dB | sens 0.00323 | yield 10.5% [9.2,11.9] | max|x| 1.12
restart 10 | eye-fit 0.53 dB | sens 0.00322 | yield 9.6% [8.3,10.9] | max|x| 1.12
restart 11 | eye-fit 0.53 dB | sens 0.00323 | yield 12.8% [11.4,14.3] | max|x| 1.12

fit gate (1.5 dB eye-band): 12/12 candidates pass
portfolio diversity: pairwise |dx| mean 0.070, max 0.243 (normalized units; ~0 would mean all restarts found the same design)

=====
RANKED PORTFOLIO (top 5) sample 1260 [var_l1_ipc]
=====
rank eye-fit dB yield % sens file
1 0.53 13.2 0.00326 design_sample_1260_var_l1_ipc_r03.npz
2 0.53 13.1 0.00324 design_sample_1260_var_l1_ipc_r05.npz
3 0.53 12.8 0.00323 design_sample_1260_var_l1_ipc_r11.npz
4 0.53 11.7 0.00322 design_sample_1260_var_l1_ipc_r08.npz
5 0.53 11.2 0.00327 design_sample_1260_var_l1_ipc_r00.npz
```

fig3: Jacobian method yield optimization

```

venv) anandshanki.amesh@anandshanki:~/mace_project_inverse_model/generative_inverse_design_of_high-speed_interconnects/open
e_model_generated.py --designs ../sandbox_v1/models/evaluation_results/generated_designs/design_sample_1260_var_L1_ipc
=====
stage 07: design_sample_1260_var_L1_ipc_top (template sim_pkg_1027, pair 1, geometry='x_local_norm')
=====

--- geometry sanity check [design_sample_1260_var_L1_ipc_top] ---
via_radius   = 12.831 mil
antipad_rad  = 21.427 mil
pitch        = 45.205 mil
TDIEL/TMET   = 181.7 total, eps_r=2.915, tan_d=0.0158
n_ports      = 12
geometry looks physical (via < antipad < pitch/2, eps_r>1).

```

Fig4: Geometry of design with $\lambda = 1.0$

Applying the original Jacobian penalty proposal did not produce any improvement to the design - it slightly reduced the sensitivity but the design still stays the same as without the yield push.

Try 3: Worst case Method (Wang, F., Lazarov, B.S. & Sigmund, O. On projection methods, convergence and robust formulations in topology optimization. *Struct Multidisc Optim* **43**, 767–784 (2011). <https://doi.org/10.1007/s00158-010-0602-y>)

```

      TMET : rel +/-3.3% (limit/3)      sigma_phys=0.08007 feat-units sigma_norm=0.0818
      PERMITTIVITY : rel +/-0.67% (limit/3)      sigma_phys=0.02606 feat-units sigma_norm=0.0406
      CONDUCTIVITY : rel +/-3.3% (limit/3)      sigma_phys=0.01423 decades sigma_norm=0.3035
      LOSS/TANGENT : rel +/-3.3% (limit/3)      sigma_phys=0.01423 decades sigma_norm=0.0351
(IPC gives ACCEPTANCE LIMITS, not process sigmas; limit/3 is a stated Cpk=1 assumption. CONDUCTIVITY entry is an engineering assum

=== YIELD TTO [wc_L1_ipc] sample 5490: 12 restarts x 150 steps
    lambda_cc=1.0 [Wang/Sigmund worst-case corners]
    spec: eye-band IL >= target-1.0 dB, RL <= -8.0 dB | tol_frac=0.1
restart 00 | eye-fit 0.69 dB | sens 0.01081 | yield 98.6% [97.9,99.0] | max|x| 0.96
restart 01 | eye-fit 0.69 dB | sens 0.01087 | yield 99.1% [98.5,99.4] | max|x| 0.96
restart 02 | eye-fit 0.54 dB | sens 0.01183 | yield 100.0% [99.8,100.0] | max|x| 0.90
restart 03 | eye-fit 0.69 dB | sens 0.01081 | yield 98.7% [98.1,99.1] | max|x| 0.96
restart 04 | eye-fit 0.54 dB | sens 0.01184 | yield 100.0% [99.8,100.0] | max|x| 0.90
restart 05 | eye-fit 0.57 dB | sens 0.01234 | yield 100.0% [99.8,100.0] | max|x| 1.51
restart 06 | eye-fit 0.69 dB | sens 0.01087 | yield 98.8% [98.2,99.2] | max|x| 0.96
restart 07 | eye-fit 0.69 dB | sens 0.01085 | yield 99.1% [98.5,99.4] | max|x| 0.96
restart 08 | eye-fit 0.69 dB | sens 0.01083 | yield 98.9% [98.3,99.2] | max|x| 0.96
restart 09 | eye-fit 0.53 dB | sens 0.01143 | yield 100.0% [99.8,100.0] | max|x| 0.88
restart 10 | eye-fit 0.54 dB | sens 0.01183 | yield 100.0% [99.8,100.0] | max|x| 0.90
restart 11 | eye-fit 0.52 dB | sens 0.01188 | yield 100.0% [99.8,100.0] | max|x| 0.97

fit gate (1.5 dB eye-band): 12/12 candidates pass
portfolio diversity: pairwise |dx| mean 0.792, max 2.692 (normalized units; ~0 would mean all restarts found the same design)

=====
RANKED PORTFOLIO (top 5) sample 5490 [wc_L1_ipc]
=====
rank eye-fit dB yield %      sens file
1      0.54    100.0    0.01183 design_sample_5490_wc_L1_ipc_r02.npz
2      0.54    100.0    0.01184 design_sample_5490_wc_L1_ipc_r04.npz
3      0.57    100.0    0.01234 design_sample_5490_wc_L1_ipc_r05.npz
4      0.53    100.0    0.01143 design_sample_5490_wc_L1_ipc_r09.npz
5      0.54    100.0    0.01183 design_sample_5490_wc_L1_ipc_r10.npz

csv      : portfolio_sample_5490_wc_L1_ipc_20260730_000503.csv
pareto   : pareto_sample_5490_wc_L1_ipc_20260730_000503.png
top design (for stage 07): design_sample_5490_wc_L1_ipc_top.npz

```

fig 5: Worst case yield optimization for sample 5490

```
(.venv) dhanushkramesh@Dhanush:~/mece_project_inverse_model/Generative_Inverse_Design_of_High-Speed_Interconnects/openEMS_sim$ python3 validate
evaluation_results/generated_designs/design_sample_5490_wc_l1_ipc_top.npz --check-only
=====
Stage 07: design_sample_5490_wc_l1_ipc_top (template sim_pkg_4511, pair 6, geometry='x_local_norm')
=====

--- geometry sanity check [design_sample_5490_wc_l1_ipc_top] ---
via_radius    = 6.016 mil
antipad_rad   = 23.690 mil
pitch         = 57.585 mil
TDIEL/TMET    = 184.1 total, eps_r=3.697, tan_d=0.0213
n_ports       = 32
geometry looks physical (via < antipad < pitch/2, eps_r>1).

[--check-only] stopping before simulation. If the geometry looks good, re-run without --check-only.
```

fig 6: geometry of best design for sample 5490

```
(.venv) dhanushkramesh@Dhanush:~/mece_project_inverse_model/Generative_Inverse_Design_of_High-Speed_Interconnects/sandbox_v1/models$ python3 validate
=====
TDIEL : rel +/-3.3% (limit/3)      sigma_phys=0.9074 feat-units sigma_norm=0.0413
TMET : rel +/-3.3% (limit/3)      sigma_phys=0.08801 feat-units sigma_norm=0.0899
PERMITTIVITY : rel +/-0.67% (limit/3) sigma_phys=0.02094 feat-units sigma_norm=0.0326
CONDUCTIVITY : rel +/-3.3% (limit/3) sigma_phys=0.01423 decades sigma_norm=0.3035
LOSSTANGENT : rel +/-3.3% (limit/3) sigma_phys=0.01423 decades sigma_norm=0.0351
(IPC gives ACCEPTANCE LIMITS, not process sigmas; limit/3 is a stated Cpk=1 assumption. CONDUCTIVITY entry is an engineering assumption with no spec or. See PartC_Yield_Plan_v3.md.)

=== YIELD TTO [wc_l1_ipc] sample 1260: 12 restarts x 150 steps
lambda_cc=1.0 [Wang/Sigmund worst-case corners]
spec: eye-band IL >= target-1.0 dB, RL <= -8.0 dB | tol frac=0.1
restart 00 | eye-fit 2.15 dB | sens 0.00334 | yield 78.6% [76.7,80.3] | max|x| 2.75
restart 01 | eye-fit 2.67 dB | sens 0.00500 | yield 80.3% [78.6,82.0] | max|x| 2.73
restart 02 | eye-fit 1.62 dB | sens 0.00202 | yield 76.6% [74.7,78.5] | max|x| 2.65
restart 03 | eye-fit 2.41 dB | sens 0.00399 | yield 80.8% [79.0,82.4] | max|x| 2.24
restart 04 | eye-fit 1.84 dB | sens 0.00247 | yield 77.3% [75.4,79.1] | max|x| 2.73
restart 05 | eye-fit 1.70 dB | sens 0.00248 | yield 79.2% [77.4,81.0] | max|x| 2.99
restart 06 | eye-fit 1.14 dB | sens 0.00279 | yield 30.1% [28.1,32.1] | max|x| 2.44
restart 07 | eye-fit 1.64 dB | sens 0.00211 | yield 76.6% [74.7,78.4] | max|x| 2.92
restart 08 | eye-fit 1.52 dB | sens 0.00207 | yield 74.8% [72.9,76.7] | max|x| 2.80
restart 09 | eye-fit 1.63 dB | sens 0.00176 | yield 71.7% [69.7,73.6] | max|x| 3.14
restart 10 | eye-fit 1.80 dB | sens 0.00283 | yield 78.0% [76.1,79.8] | max|x| 2.57
restart 11 | eye-fit 1.56 dB | sens 0.00223 | yield 77.6% [75.7,79.4] | max|x| 2.74

fit gate (1.5 dB eye-band): 1/12 candidates pass

=====
RANKED PORTFOLIO (top 5) sample 1260 [wc_l1_ipc]
=====
rank eye-fit dB yield % sens file
1 1.14 30.1 0.00279 design_sample_1260_wc_l1_ipc_r06.npz

csv : portfolio_sample_1260_wc_l1_ipc_20260730_000900.csv
pareto : pareto_sample_1260_wc_l1_ipc_20260730_000900.png
top design (for stage 07): design_sample_1260_wc_l1_ipc_top.npz
(.venv) dhanushkramesh@Dhanush:~/mece_project_inverse_model/Generative_Inverse_Design_of_High-Speed_Interconnects/sandbox_v1/models$
```

fig 7: Worst case method for sample 1260 where the yield is better but the fit is sacrificed

I tried the worst case yield aware optimization on the validation set - which is inspired from the worst case penalty from Wang et al. the optimizer nudges each geometry dimension to its $\pm 1\sigma$ manufacturing extreme (the ± 0.5 mil drill wobble, $\pm 3\%$ material variation from the IPC tolerance model) and ensures that even the *worst* of these perturbed corners still meets spec. this will result in the design that is robust to the manufacturing tolerances. In this method few samples had a trade off in the actual fit with the target with 2.5dB - but with 80% yield and few samples had perfect match and with 98 to 100% yield. It totally depends on the complexity. This study was useful to find whether the robust manufacturing design exists for a particular target and the trade off between the manufacturability (yield) and the fit. I am yet to run the openEMS

simulation for the validation samples run though worst-case yield optimization method. i will share the eye diagram once the run is complete in openEMS.